KCOR Supplementary Information (SI)

This document provides supplementary material supporting the KCOR methodology described in the main manuscript, including extended derivations, simulation studies, robustness analyses, and additional empirical results.

S1. Overview

This SI is organized as follows:

- **S1**: Overview
- **S2**: Extended diagnostics and failure modes
- **S3**: Positive controls (injected harm/benefit)
- **S4**: Control-test specifications and simulation parameters
- **S5**: Additional figures and diagnostics
- **S6**: Extended Czech empirical application / illustrative registry analysis

S2. Extended diagnostics and failure modes

This section describes the **observable diagnostics and failure modes** associated with the revised KCOR working assumptions and the corresponding identifiability criteria. KCOR is designed to **fail transparently rather than silently**: when an assumption is violated, the resulting lack of identifiability or model stress manifests through explicit diagnostic signals rather than spurious estimates. In particular, non-multiplicative or strongly time-varying components within quiet windows can distort curvature and obscure depletion geometry, and heterogeneous multiplicative effects across cohorts can perturb depletion dynamics; such cases are expected to trigger diagnostic instability or non-identifiability.

The KCOR framework separates **working assumptions**, **empirical diagnostics**, and **identifiability criteria**; these are summarized below in Tables S1–S3.

Table S1: KCOR working assumptions.

Assumption	Description	Role in KCOR
A1 Cohort stability	Cohorts are fixed at enrollment with no post-enrollment switching or informative censoring.	Ensures cumulative hazards are comparable over follow-up
A2 Shared external hazard environment	Cohorts experience the same background hazard over the comparison window, except for explicitly modeled extensions.	Prevents confounding by cohort-specific shocks

Assumption	Description	Role in KCOR
A3 Enrollment-time frailty	Selection operates through time-invariant latent heterogeneity summarized by cohort-specific enrollment-time frailty variance $\theta_{0,d}$, estimated separately for each cohort. Under minimal aggregated data, $\theta_{0,d}$ is interpreted via curvature in cumulative-hazard space; constant multiplicative effects are not separately identifiable from baseline scale. Curvature-based identification is invariant to multiplicative scaling that is common or approximately common across cohorts, while heterogeneous multiplicative effects and non-multiplicative contamination are assessed via diagnostics (main text §2.4.4).	Defines the parameter targeted by the estimator
A4 Adequacy of gamma frailty	Gamma frailty provides a reasonable approximation to observed depletion geometry.	Allows tractable inversion and normalization
A5 Adequacy of Gompertz baseline	A fixed Gompertz age slope is a reasonable working approximation over the estimation horizon.	Anchors the structured baseline used to identify $\theta_{0,d}$
A6 Multi-window quiet-period validity	Prespecified quiet windows exist in which depletion can be identified without dominant external shocks.	Permits pooled identification across windows within each cohort
A7 Structured offset additivity	When the delta path is used, wave effects are additive in cumulative-hazard space and persist forward after the wave ends.	Permits alignment of quiet windows through δ_i and $\Delta(t)$
A8 Optional NPH exponent model identifiability	When the optional NPH module is used, excess hazard is reasonably represented by a shared-amplitude frailty-dependent amplification model with a global exponent α; α is localized using cross-cohort structure together with **externally supplied VE or other prespecified intervention-effect scale** not learned from the same wave-period contrasts; and the cumulative correction is applied **after** gamma-frailty inversion (main text §2.7).	Supports optional post-inversion NPH correction

Table S2: Empirical diagnostics associated with KCOR assumptions.

Diagnostic	Description	Observable signal
Skip-week sensitivity	Exclude early post-enrollment weeks subject to dynamic selection.	Stable fitted $\theta_{0,d}$ under varying skip weeks
Seed-fit quality	Evaluate the nearest-window Gompertz seed fit.	Small residuals and plausible $\hat{k}_d$ / $\hat{\theta}^{(0)}_{0,d}$
Multi-window consistency	Refit after omitting or perturbing individual quiet windows.	Stable $\hat{\theta}_{0,d}$ and similar normalized trajectories
Delta plausibility	Check reconstructed offsets and failure flags.	Positive / coherent offsets or transparent fallback
Post-normalization linearity	Assess curvature removal in cumulative-hazard space.	Approximate linearity after normalization within quiet windows as a diagnostic check under the working model
Residual structure	Examine residuals in hazard space.	No systematic time-structure within or across windows
KCOR(t) stability	Inspect KCOR trajectories following anchoring.	Stabilization rather than drift, interpreted diagnostically rather than as proof
Optional NPH diagnostics	Compare pairwise and collapse estimators, theta propagation scales, excess-handling choices, post-inversion correction diagnostics, and with/without-module results when relevant.	Estimator agreement, non-flat objectives, and limited dependence on reasonable NPH settings

Empirical illustration of the "Quiet-window perturbation" diagnostic is provided in Figure S5, which scans monthly-shifted 12-month windows in Czech registry data.

Table S3: Identifiability criteria governing KCOR interpretation.

Criterion	Condition	Consequence if violated
I1 Diagnostic sufficiency	All required diagnostics pass.	KCOR interpretable
I2 Curvature sufficiency	Mortality geometry contains enough curvature to distinguish k_d from $\theta_{0,d}$, with diagnostics supporting curvature-based identification (including adequacy of common multiplicative scaling across cohorts where relevant and absence of dominant non-multiplicative contamination in the quiet-window regime).	$\theta_{0,d}$ weakly identified
I3 Multi-window coherence	Quiet windows remain consistent after alignment.	Interpretation limited
I4 Delta applicability	Offset reconstruction is coherent when the delta path is used.	Fall back or treat as non-identifiable
I5 Anchoring validity	Post-normalization behavior is stable in the reference quiet window.	Anchoring invalid

Criterion	Condition	Consequence if violated
I6 NPH signal sufficiency	When the optional NPH module is used, cross-cohort differences in depletion geometry and excess hazard are strong enough to localize α jointly with a prespecified **externally supplied VE or other prespecified intervention-effect scale**.	α weakly identified; objectives flatten or estimators disagree
I7 Conservative failure rule	Any failure → diagnostics indicate non-identifiability.	Analysis treated as not identified; results not reported

When diagnostics indicate non-identifiability, the analysis is treated as not identified and results are not reported; this does not invalidate the KCOR estimator itself. In minimal aggregated data, identifiability of $\theta_{0,d}$ from curvature remains conditional on the working model rather than assumption-free: additive or sharply time-varying components can distort curvature, and heterogeneous multiplicative effects across cohorts can perturb depletion paths, with diagnostics intended to surface these failure modes. When the optional NPH module is used, identifiability of α is likewise conditional: it depends on cross-cohort separation in depletion geometry, stable excess-hazard measurement, and a prespecified **externally supplied VE or other prespecified intervention-effect scale** (main text §2.7.2); α is not separately identifiable from cohort-specific multiplicative effects under minimal data if that scale is not fixed externally (main text §5.4). The common external intensity is not directly observed.

S2.1 Optional NPH exponent model: estimator and failure signatures

The optional NPH module extends KCOR only for prespecified periods in which an external hazard may interact with frailty in a non-proportional manner. Under the working model, excess hazard for cohort d is represented as

$$h_{\text{excess},d}(t) = A(t)\, F_d(t;\alpha), \qquad F_d(t;\alpha) = E_d[z^\alpha \mid t],$$

where $A(t)$ is an unknown common amplitude and the exponent α is treated as global across cohorts within the analyzed period. When $\alpha = 1$, amplification is proportional to baseline risk and the NPH module is effectively inactive. The exponent α should be interpreted as a model-calibrated summary of frailty-dependent amplification under the working model, not as a uniquely identified biological or mechanistic constant.

Because $A(t)$ is common across cohorts, absolute excess hazards are not directly comparable. However, cross-cohort ratios eliminate the unknown common amplitude:

$$\frac{h_{\text{excess},i}(t)}{h_{\text{excess},j}(t)} = \frac{F_i(t;\alpha)}{F_j(t;\alpha)}.$$

Post-inversion correction. Applying the NPH correction after gamma-frailty inversion ensures wave-period excess is measured relative to a frailty-neutral baseline in cumulative-hazard space; correcting raw hazards would confound depletion geometry with amplification (main text §2.7.3). In discrete weekly bins, a Gompertz reference path is anchored at wave entry, only strictly positive excess above that path is divided by $F_d(t;\alpha)$, and the corrected cumulative hazard is enforced to be non-decreasing.

Identification with fixed external scale. Under minimal aggregated data, α is not separately identifiable from cohort-specific multiplicative intervention effects unless **externally supplied VE or other prespecified intervention-effect scale** is fixed and not estimated from the same wave-period contrasts used to localize α (main text §2.7.2). Given a fixed scale x relating frailty-neutral excess hazards between cohorts A and B,

$$\frac{\Delta h_A^{\text{obs}}(t)}{\Delta h_B^{\text{obs}}(t)} = (1-x)\,\frac{F_A(t;\alpha)}{F_B(t;\alpha)}.$$

With **externally supplied VE or other prespecified intervention-effect scale** fixed as in main text §2.7.2, the operational **pairwise estimator** minimizes discrepancies between observed log excess-hazard ratios and $\log(1-x) + \log F_i(t;\alpha) - \log F_j(t;\alpha)$, in the same spirit as

$$\sum_{i<j,t} \left[\log e_i(t) - \log e_j(t) - \left(\log F_i(t;\alpha) - \log F_j(t;\alpha)\right)\right]^2,$$

but with the appropriate $\log(1-x)$ offset absorbed into the target or the residual as implemented in the reference code. The **collapse estimator** minimizes

$$\mathrm{Var}_d\left[\log e_d(t) - \log F_d(t;\alpha)\right]$$

under the same fixed-scale restriction.

Agreement between the two estimators is treated as supportive of identification under the working model; disagreement is diagnostic rather than something to be averaged away. Additional robustness checks vary the excess-hazard handling rule, baseline anchor choice, and theta propagation scale.

Failure of NPH identification manifests through observable signatures: flat objective curves, boundary-seeking estimates, strong dependence on arbitrary settings, or disagreement between the pairwise and collapse estimators. These patterns are treated as evidence of weak signal or model misspecification rather than as valid estimates of α.

S3. Positive controls

S3.1 Construction of injected effects

The effect window is a simulation construct used solely for positive-control validation and does not represent a real-world intervention period or biological effect window.

Positive controls are constructed by starting from a negative-control dataset and injecting a known effect into the data-generating process for one cohort, for example by multiplying the *baseline* hazard by a constant factor r over a prespecified interval:

$$h_{0,\text{treated}}(t) = r \cdot h_{0,\text{control}}(t) \quad \text{for } t \in [t_1, t_2], \tag{1}$$

with $r > 1$ for harm and $0 < r < 1$ for benefit.

After gamma-frailty normalization (inversion), KCOR should deviate from 1 in the correct direction and with magnitude broadly consistent with the injected effect (up to discretization and sampling noise). Figure S1 and main-text Table 9 illustrate this behavior.

S4. Control-test specifications and simulation parameters

S4.0 Summary tables for control-test and simulation parameters

Table S4: Summary of control-test and simulation parameters referenced in Sections S4.2–S4.6. Values shown are representative defaults or grids used to probe the revised estimator; exact scripts and run settings are version-controlled in the repository.

Section	Item	Parameter	Value	Notes
S4.2.1	Synthetic validation	Data source / scripts	Repository simulation scripts	Used for θ_0 recovery and null-behavior checks

Section	Item	Parameter	Value	Notes
S4.2.1	Synthetic validation	Frailty heterogeneity grid	Prespecified varying grid	Includes weak- and strong-curvature regimes
S4.2.1	Synthetic validation	Baseline model in estimator	Gompertz with fixed γ	Aligns with main-text estimator
S4.2.1	Synthetic validation	Time horizon	Multi-year follow-up	Sufficient to probe identifiability
S4.2.2	Empirical negative control	Data source	Czech admin registry data (KCOR_CMR)	Aggregated cohorts
S4.2.2	Empirical negative control	Construction	Age strata remapped to pseudo-doses	Preserves a pseudo-null by design
S4.3	Positive control	Effect multiplier	$r = 1.2$ harm; $r = 0.8$ benefit	Injected after the same preprocessing pipeline
S4.3	Positive control	Frailty target	Enrollment-time $\theta_{0,d}$	Estimated separately within each cohort, not fixed in the estimator
S4.4	Sensitivity analysis	Perturbations	Skip weeks, quiet-window placement, omitted windows, anchor choice, excess handling, theta propagation scale, optional NPH settings	Probes robustness of the revised estimator
S4.5	Adversarial selection geometry	Frailty distribution	Tail-mixture versus mid-quantile sampling	Tests robustness to non-gamma geometry
S4.6	Joint frailty + effect	DGP	Frailty heterogeneity plus injected effect	Tests separability of depletion and effect windows

S4.1 Reference implementation and default operational settings

Table S5: Reference implementation and default operational settings.

Component	Setting	Default value	Notes
Cohort construction	Cohort indexing	Enrollment period × YearOfBirth group × Dose; plus all-ages cohort (YearOfBirth = -2)	Implementation detail
Quiet-period selection	Quiet windows	Prespecified ISO-week intervals spanning follow-up	Used jointly after alignment, not one at a time

Component	Setting	Default value	Notes
Early-period stabilization (dynamic HVE)	SKIP_WEEKS	2	Weeks $t <$ SKIP_WEEKS are excluded from accumulation; time is then rebased
Gompertz slope	\gamma	Fixed implementation value	Same value used throughout the estimator
Frailty estimation	Fit method	Seed Gompertz fit + delta iteration + pooled refit	Estimates $(k_d, \theta_{0,d})$ separately within each cohort in the main path
Anchoring	NORMALIZATION_WEEKS	4	Reference window for anchored KCOR plots
NPH module	Optional NPH exponent model	Off by default	Enabled only when cross-cohort NPH signal is plausible and diagnostics support identification of α

S4.2 Negative controls

Negative controls are used to evaluate the behavior of KCOR under settings where the true effect is known to be null, while allowing substantial heterogeneity in baseline risk and selection-induced depletion. Two complementary classes of negative controls are considered: (i) fully synthetic simulations used to assess both θ_0 recovery and null KCOR behavior, and (ii) empirical registry-based constructions that preserve a pseudo-null by repurposing age strata as pseudo-exposures without selective sampling. Together, these controls assess whether KCOR remains stable in the presence of non-proportional hazards arising from selection rather than treatment.

S4.2.1 Synthetic negative control: gamma-frailty null

The synthetic negative control (Figure S3) is a fully specified simulation designed to induce **strong selection-induced depletion curvature under a true null effect**. Under the revised estimator, this synthetic setting is used not only to ask whether KCOR remains near 1, but also whether the estimator recovers the cohort-specific enrollment-time frailty parameters governing the simulated geometry and aligns quiet windows coherently. Near-1 behavior here is interpreted as a diagnostic implication under the working model, not as stand-alone proof that the estimator is correct.

Parameter values and scripts are summarized in Table S4.

Both cohorts share identical per-frailty-group death probabilities; only the mixture weights differ. This induces different cohort-level curvature under the null.

Figure S2 provides a compact summary of KCOR bias as a function of frailty variance under the same synthetic-null grid used in main-text Table 7.

S4.2.2 Empirical negative control: age-shift construction

The empirical negative control (main-text Figure 6 and SI Figure S9) repurposes registry cohorts to create a **pseudo-null comparison** while inducing large baseline hazard differences via 10–20 year age shifts. Because these are full-population strata rather than selectively sampled subcohorts, selection-induced depletion is reduced and the fitted $\theta_{0,d}$ values are expected to be small or weakly identified. The goal here is to test end-to-end KCOR behavior, not to use the empirical geometry as proof that all confounding has been removed.

Parameter values and scripts are summarized in Table S4.

This construction ensures that dose comparisons are within the same underlying vaccination category, preserving a pseudo-null while inducing 10–20 year age differences.

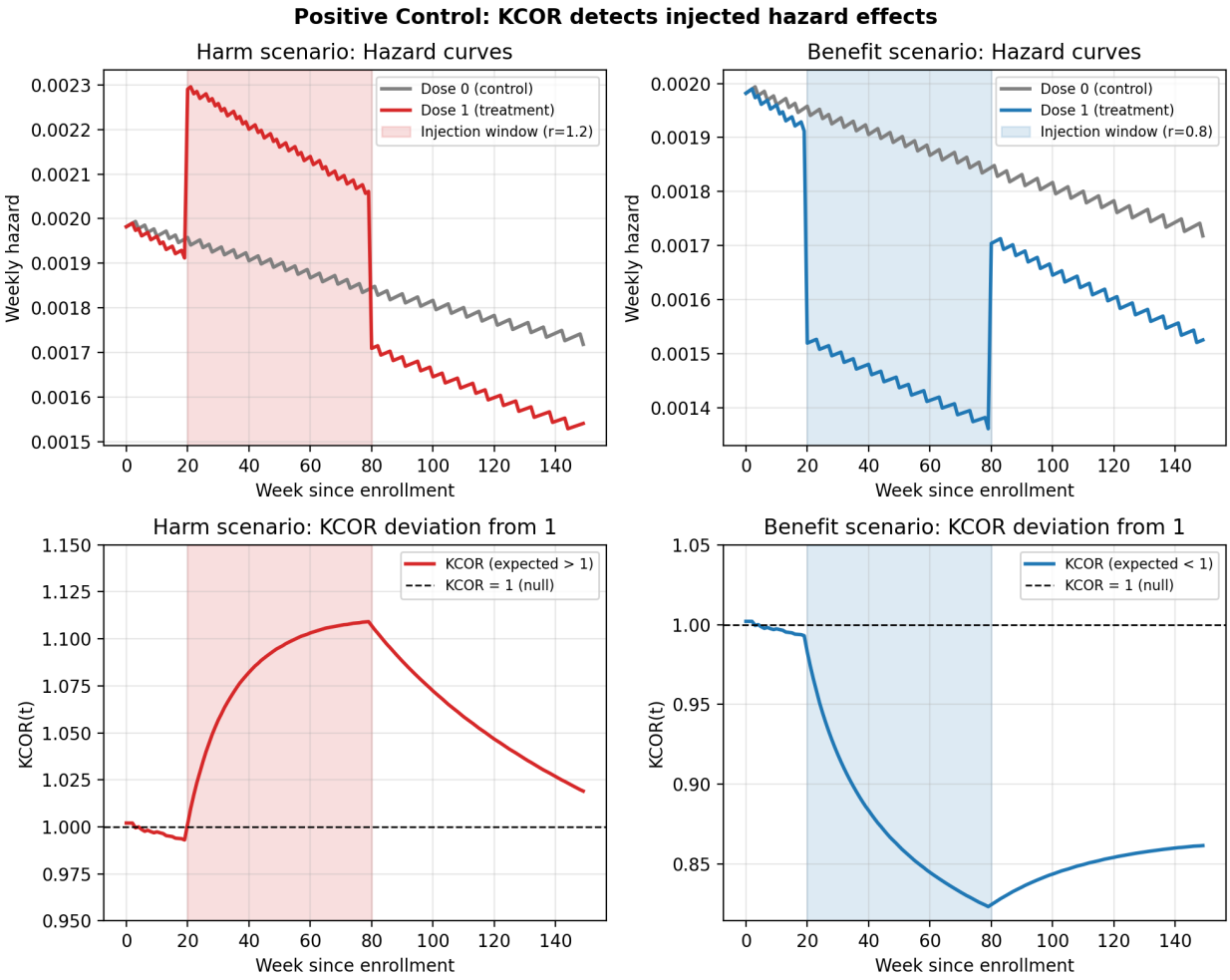


Figure S1: Positive control validation: KCOR tracks injected effects in the expected direction. Left panels show harm scenario (r=1.2), right panels show benefit scenario (r=0.8). Top row displays cohort hazard curves with effect window shaded. Bottom row shows KCOR(t) deviating from 1.0 in the expected direction during the effect window. Uncertainty bands (95% bootstrap intervals; aggregated cohort–time resampling) are shown. X-axis units are weeks since enrollment.

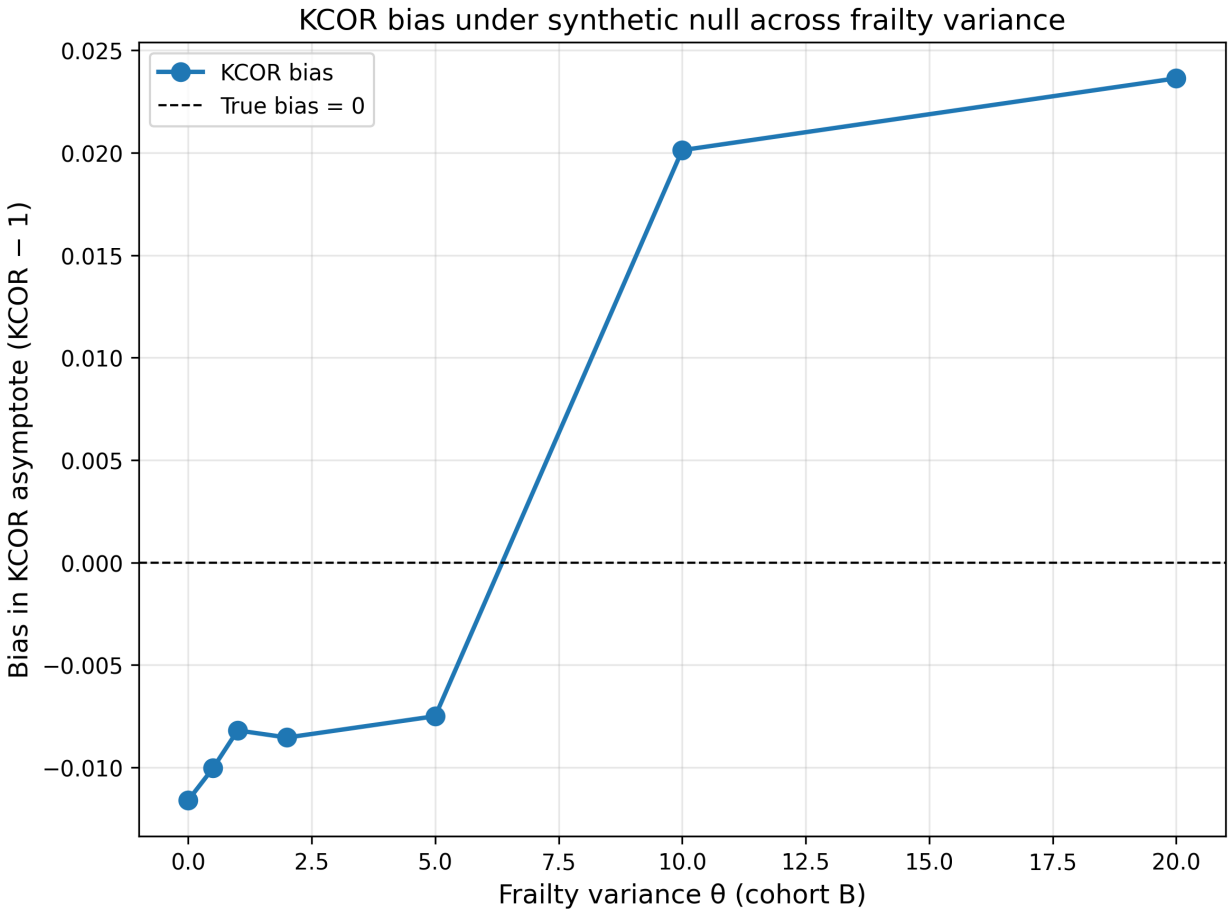


Figure S2: Simulated-null summary: KCOR bias as a function of enrollment-time frailty variance θ_0. Bias is defined as $\mathrm{KCOR}_{\mathrm{asymptote}} - 1$ at the end of follow-up in the synthetic-null grid (no treatment effect), reflecting cumulative deviation under the null rather than instantaneous hazard bias. This figure is a diagnostic summary under the working model, not proof that the model is true. Points show single-run estimates from the grid; no error bars are shown.

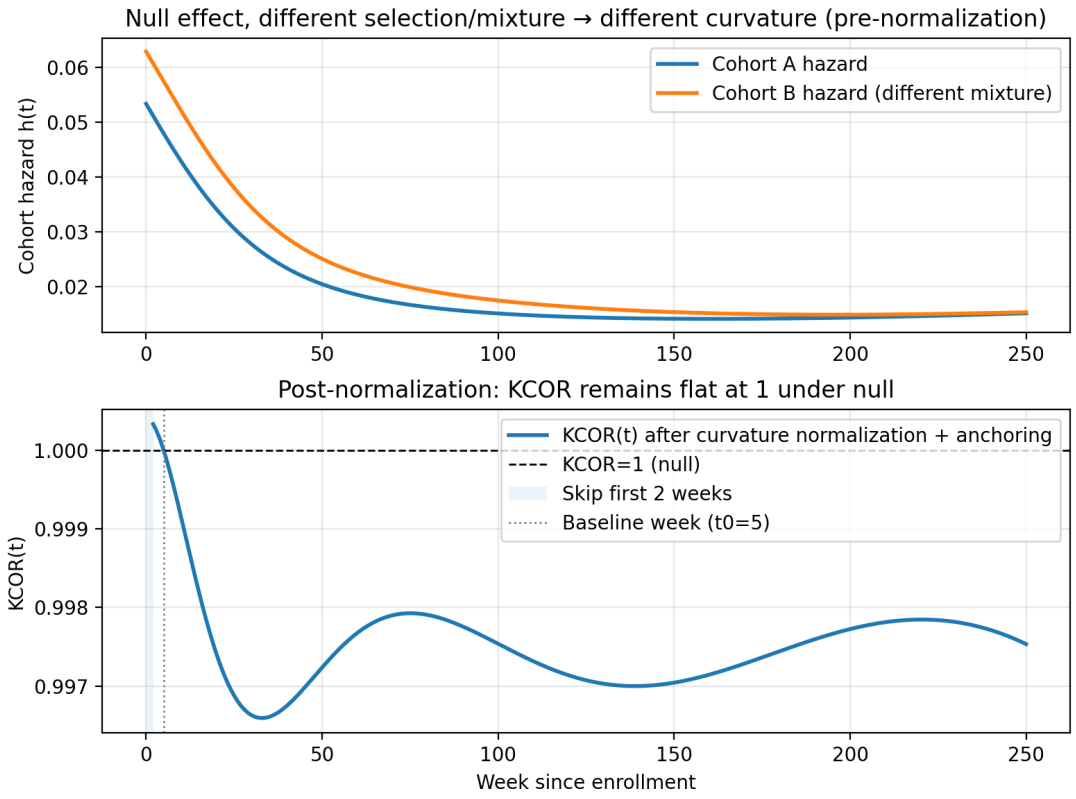


Figure S3: Synthetic negative control under strong selection (different curvature) but no effect: KCOR(t) stays near 1 under the synthetic null. Top panel shows cohort hazards with different frailty-mixture weights inducing different curvature. Bottom panel shows KCOR(t) remaining near 1.0 after normalization, which is the diagnostic behavior expected under the working model rather than proof that all confounding has been removed. Uncertainty bands (95% bootstrap intervals; aggregated cohort–time resampling) are shown.

This contrasts with the synthetic negative control (Section S4.2.1), where strong, deliberately induced frailty heterogeneity requires gamma-frailty normalization to recover the null.

S4.3 Positive control: injected effect

Positive controls are used to verify that KCOR responds appropriately when a true effect is present. Starting from a negative-control simulation with no treatment effect, a known multiplicative hazard shift is injected into one cohort over a prespecified time window. This construction allows direct assessment of whether the same preprocessing and $\theta_{0,d}$ estimation pipeline detects both the direction and timing of the injected effect while remaining stable outside the effect window.

Parameter values and scripts are summarized in Table S4.

The injection multiplies the treatment cohort's baseline hazard by factor r during the effect window, while leaving the control cohort unchanged.

S4.4 Sensitivity analysis parameters

Sensitivity analyses evaluate the robustness of KCOR estimates to reasonable variation in analysis choices that do not alter the underlying data-generating process. Skip weeks, quiet-window placement, omission of individual quiet windows, and optional NPH settings are perturbed over a prespecified range while holding all other parameters fixed. For the optional NPH module, this robustness menu includes baseline anchor choice, excess-hazard handling, theta propagation scale, and agreement between the pairwise and collapse estimators. These analyses assess whether KCOR behavior is stable to tuning choices that primarily affect identification rather than cohort composition.

Parameter values and scripts are summarized in Table S4.

Output grids show KCOR(t) values for each parameter combination.

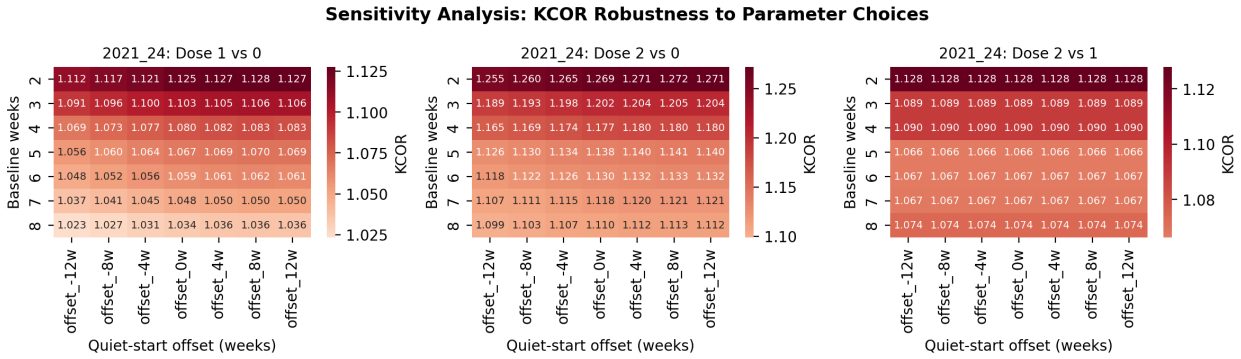


Figure S4: Sensitivity analysis summary showing KCOR(t) values across parameter grid. Heatmaps display KCOR(t) estimates for different combinations of baseline weeks (rows) and quiet-window start offsets (columns). Across all comparisons, KCOR(t) varies smoothly and modestly across a wide range of quiet-start offsets and baseline window lengths, with no qualitative changes in sign or magnitude, indicating robustness to reasonable parameter choices. All panels use a unified color scale centered at 1.0 to enable direct visual comparison across dose comparisons.

S4.5 Tail-sampling / bimodal selection (adversarial selection geometry)

This adversarial simulation evaluates KCOR under extreme but controlled violations of typical cohort-selection geometry. Two cohorts are constructed to share identical mean frailty while differing sharply in how risk is distributed, using mid-quantile sampling versus a low/high-tail mixture. This setting stress-tests whether the revised $\theta_{0,d}$ estimator and depletion normalization remain effective when frailty heterogeneity is concentrated in the tails rather than smoothly distributed.

- **Mid-sampled cohort**: frailty restricted to central quantiles (e.g., 25th–75th percentile) and renormalized to mean 1.
- **Tail-sampled cohort**: mixture of low and high tails (e.g., 0–15th and 85th–100th percentiles) with mixture weights chosen to yield mean 1.

Parameter values and scripts are summarized in Table S4.

Both cohorts share the same baseline hazard $h_0(t)$ and have no treatment effect (negative-control version). Positive-control versions are also generated by applying a known hazard multiplier in a prespecified window. The evaluation includes (i) KCOR drift, (ii) fit quality under the revised estimator, (iii) post-normalization linearity, and (iv) parameter stability under window perturbation.

S4.6 Joint frailty and treatment-effect simulation

This simulation evaluates KCOR under conditions in which **both selection-induced depletion (frailty heterogeneity)** and a **true treatment effect (harm or benefit)** are present simultaneously. The purpose is to assess whether KCOR can (i) correctly identify and neutralize frailty-driven curvature using aligned quiet windows and (ii) detect a true treatment effect outside that period without confounding the two mechanisms.

This joint simulation combines the selection-induced depletion mechanisms examined in Sections S4.2 and S4.5 with the injected-effect framework of Section S4.3.

Parameter values and scripts for this joint simulation are summarized in Table S4.

Design

Two fixed cohorts are generated with identical baseline hazards but differing enrollment-time frailty variance. Individual hazards are multiplicatively scaled by a latent frailty term drawn from a gamma distribution with unit mean and cohort-specific variance. A treatment effect is then injected over a prespecified time window that does not overlap the quiet windows used for frailty identification.

Formally, individual hazards are generated as

$$h_i(t) = z_i \cdot h_0(t) \cdot r(t). \tag{2}$$

where z_i is individual frailty, $h_0(t)$ is a shared baseline hazard, and $r(t)$ is a time-localized multiplicative treatment effect applied to one cohort only.

S5. Additional figures and diagnostics

This section provides diagnostic outputs and evaluation criteria for the simulations and control-test specifications defined in Section S4.

S5.1 Fit diagnostics

For each cohort d, the revised fit produces diagnostic outputs including:

- **RMSE in hazard space**: Root mean squared error between observed and model-predicted hazards over the pooled quiet-window fit set.
- **Fitted parameters**: baseline scale parameter and cohort-specific enrollment-time frailty variance $\theta_{0,d}$.
- **Number of fit points**: pooled observations across quiet windows.
- **Offset diagnostics**: number, sign, and stability of reconstructed δ_i terms when the delta path is used.

When uncertainty bands or bootstrap intervals are reported in this supplement, they are computed using an aggregated-data bootstrap at the cohort–time level (resampling event counts and risk-set sizes within time bins/strata), not by resampling individuals.

Example diagnostic output from the reference implementation:

```
KCOR_FIT,EnrollmentDate=2021_24,YoB=1950,Dose=0,
  k_hat=4.29e-03,theta0_hat=8.02e-01,
  RMSE_h=3.37e-03,n_obs=97,delta_count=3,success=1
```

S5.2 Residual analysis

Fit residuals should be examined in hazard space. Define residuals:

$$r_d(t) = h_d^{\mathrm{adj}}(t) - \frac{\hat{k}_d e^{\gamma t_{\mathrm{rebased}}}}{1 + \hat{\theta}_{0,d}\left(H_{\mathrm{gom},d}(t_{\mathrm{rebased}}) + \Delta_d(t)\right)}.$$

- **Systematic patterns**: Residuals should be approximately random around zero. Systematic structure suggests model inadequacy.
- **Outliers**: Individual weeks with large residuals may indicate data quality issues or external shocks.
- **Autocorrelation**: Strong autocorrelation in residuals suggests the model is missing time-varying structure.

S5.3 Parameter stability checks

Robustness of fitted parameters is assessed by:

- **Quiet-window perturbation**: Shift the quiet-window start/end by ±4 weeks and re-fit. Stable parameters should vary modestly and preserve the same qualitative interpretation.
- **Window omission**: Drop one quiet window at a time and verify that $\hat{\theta}_{0,d}$ and KCOR remain coherent.
- **Skip-weeks sensitivity**: Vary `SKIP_WEEKS` and verify KCOR trajectories remain qualitatively similar.
- **Gompertz sensitivity**: Check whether modest perturbations of γ materially alter interpretation.
- **Optional NPH sensitivity**: In epidemic-wave applications, compare results with and without the module, across plausible anchor and excess-handling choices, across theta propagation scales, and across the pairwise and collapse estimators.

S5.4 Quiet-window perturbation scan (empirical Czech data)

To assess whether diagnostically valid quiet windows are rare or fragile in applied registry data, we performed a systematic scan of quiet-window placements in the Czech Republic 2021–2024 cohort (enrollment ISO week 2021-24). We re-estimated the revised frailty parameters $(\hat{k}_d, \hat{\theta}_{0,d})$ across a sequence of overlapping quiet windows of fixed duration, holding all other settings fixed.

Specifically, we considered 12-month windows beginning at successive monthly offsets between April 2022 and April 2023 (inclusive). For each window placement, parameters were estimated using the same revised estimator and diagnostic pass/fail criteria used elsewhere in this SI. Results are shown for two birth-decade strata (1940s and 1950s), and for dose groups $d \in \{0, 1, 2\}$.

Figure S5 shows fitted $\hat{\theta}_{0,d}$ as a function of quiet-window midpoint date. The purpose of this scan is technical rather than substantive: it asks whether the estimator remains coherent across nearby quiet-window choices and whether failures are surfaced explicitly. Open markers denote windows failing diagnostics (treated as non-identifiable and not interpreted).

S5.5 Quiet-window overlay plots

Overlaying prespecified quiet windows on hazard and cumulative-hazard time series plots provides a visual diagnostic of window placement relative to mortality dynamics. The fit windows should:

- Avoid major epidemic waves or external mortality shocks
- Contain sufficient event counts for stable estimation
- Be interpretable after rebasing and, when relevant, after offset alignment

Visual inspection of quiet-window placement relative to mortality dynamics remains an essential diagnostic step, but it should be interpreted jointly with the formal multi-window and offset diagnostics described above.

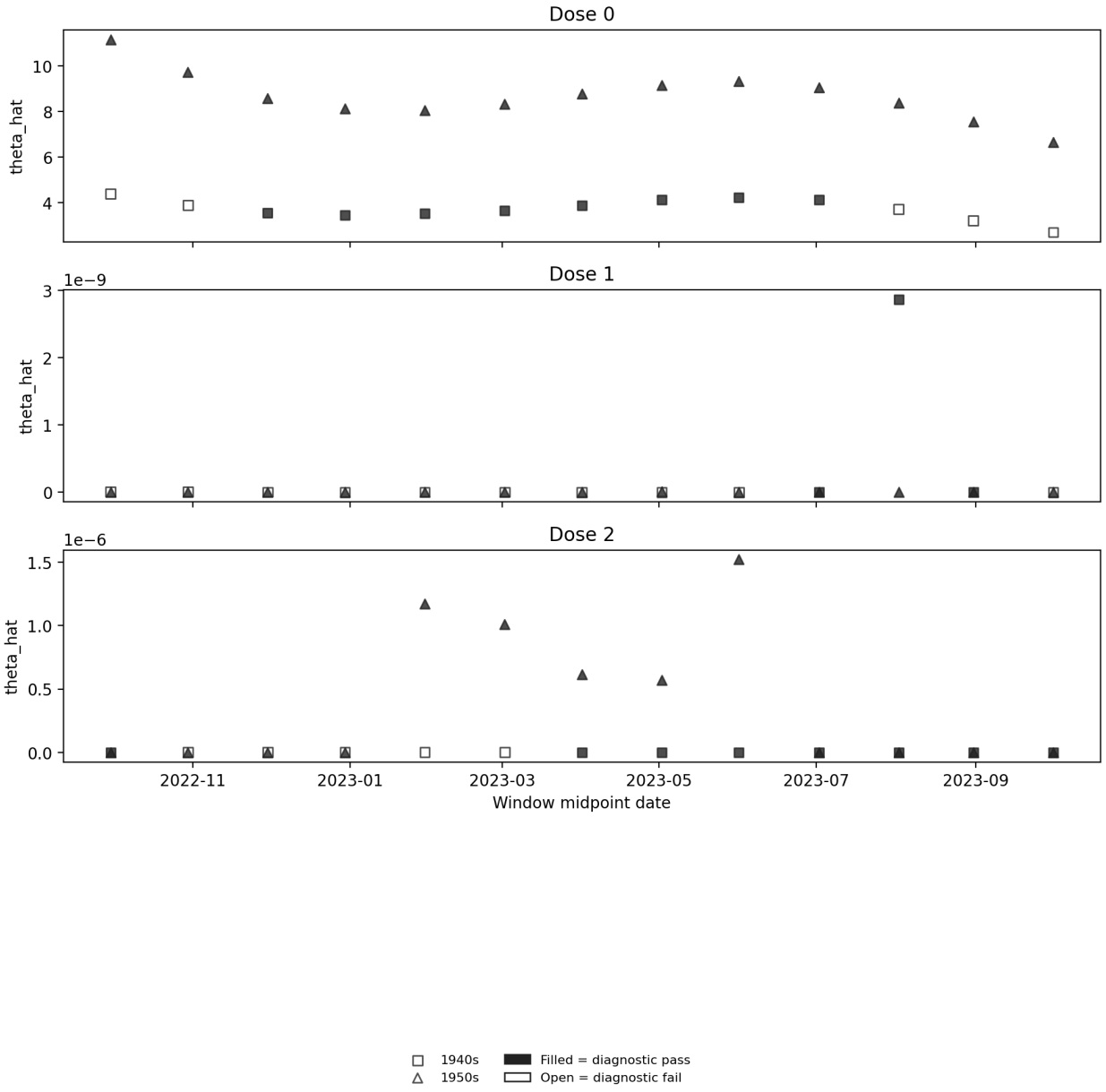


Figure S5: Quiet-window robustness scan: fitted enrollment-time frailty variance $\hat{\theta}_{0,d}$ vs. quiet-window midpoint date for the Czech 2021–2024 enrollment cohort, using 12-month windows shifted monthly from April 2022 through April 2023. Marker shape denotes birth decade (1940s, 1950s). Filled markers indicate diagnostic pass; open markers indicate diagnostic failure. The figure is intended as a technical stability scan of the estimator rather than as substantive evidence for any one cohort narrative.

S5.6 Robustness to age stratification

This subsection examines KCOR(t) under narrow age stratification by repeating the same fixed-cohort comparison in three single birth-year cohorts spanning advanced ages (1930, 1940, 1950). Across these strata, the trajectories remain qualitatively stable after depletion normalization, which is consistent with the observed behavior not being driven solely by age aggregation.

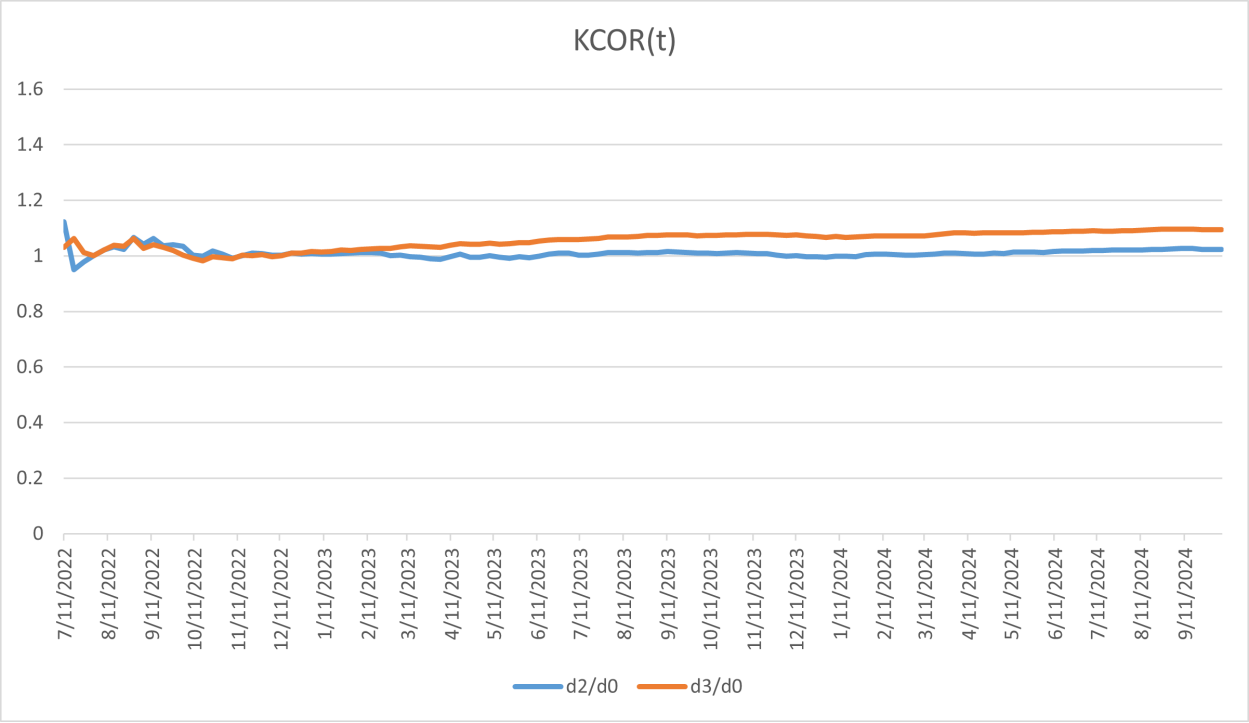


Figure S6: Birth-year cohort 1930: KCOR(t) trajectories comparing dose 2 and dose 3 to dose 0 for cohorts enrolled in ISO week 2022-26 and evaluated over calendar year 2023. KCOR curves are anchored at t_0 = 4 weeks (i.e., plotted as KCOR(t; t_0)). This figure is presented as an illustrative application demonstrating estimator behavior on registry data and does not support causal inference. X-axis units are weeks since enrollment.

Additional empirical negative-control variant (20-year age shift).
For completeness, the more extreme 20-year age-shift negative control referenced in the main text is included:

S6. Extended Czech empirical application

S6.1 Empirical application with diagnostic validation: Czech Republic national registry mortality data

The Czech results do not validate KCOR; they represent an application that satisfies all pre-specified diagnostic criteria. Substantive implications follow only if the identification assumptions hold. Throughout this subsection, observed divergences are interpreted strictly as properties of the estimator under real-world selection, not as intervention effects.

Unless otherwise noted, KCOR curves in the Czech analyses are shown anchored at t_0 = 4 weeks for interpretability.

S6.1.1 Illustrative empirical context: COVID-19 mortality data

The COVID-19 vaccination period provides a natural empirical regime characterized by strong selection heterogeneity and non-proportional hazards, making it a useful illustration for the KCOR framework. During this period, vaccine uptake was voluntary, rapidly time-varying, and correlated with baseline health status, creating clear examples of selection-induced non-proportional hazards. The Czech Republic national mortality registry data

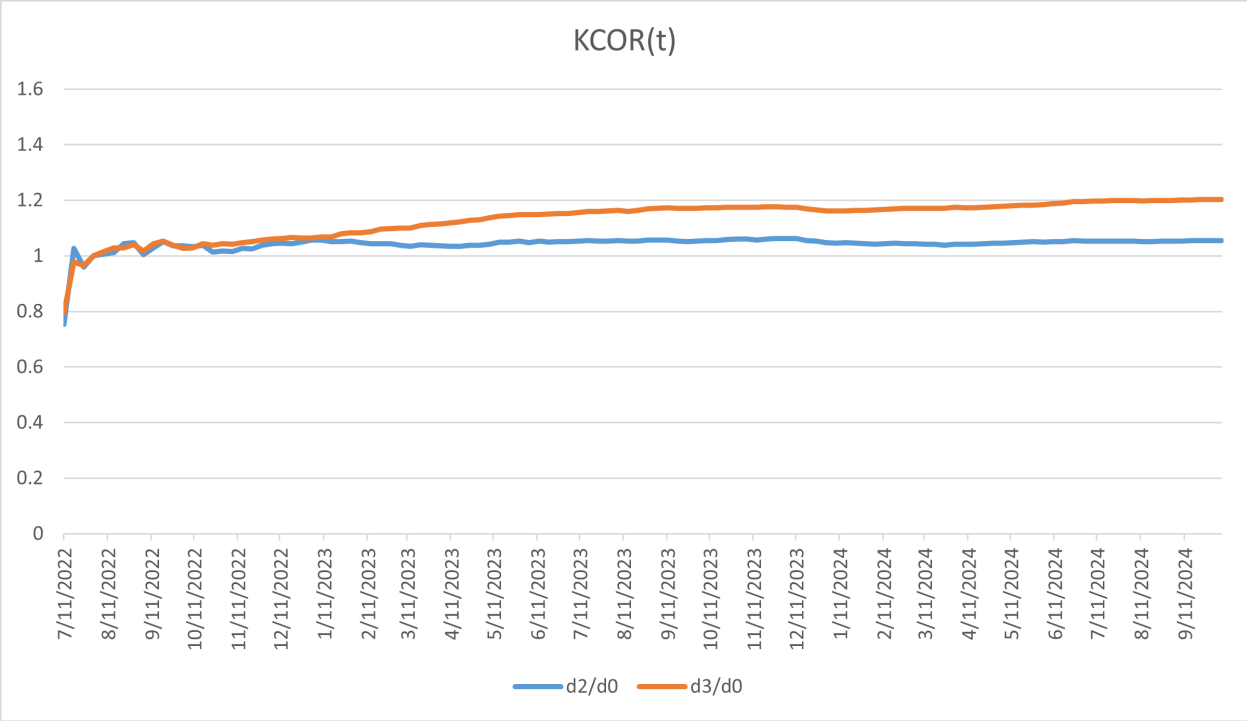


Figure S7: Birth-year cohort 1940: KCOR(t) trajectories comparing dose 2 and dose 3 to dose 0 for cohorts enrolled in ISO week 2022-26 and evaluated over calendar year 2023. KCOR curves are anchored at t_0 = 4 weeks (i.e., plotted as KCOR$(t; t_0)$). This figure is presented as an illustrative application demonstrating estimator behavior on registry data and does not support causal inference. X-axis units are weeks since enrollment.

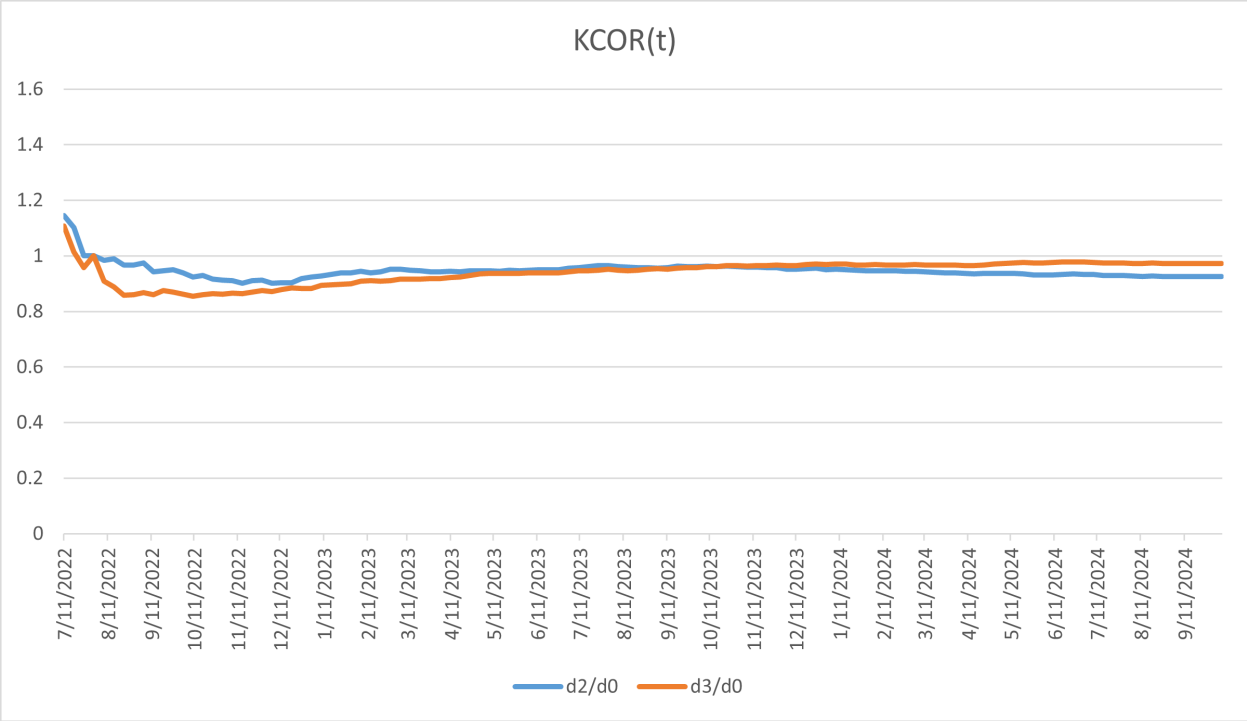


Figure S8: Birth-year cohort 1950: KCOR(t) trajectories comparing dose 2 and dose 3 to dose 0 for cohorts enrolled in ISO week 2022-26 and evaluated over calendar year 2023. KCOR curves are anchored at t_0 = 4 weeks (i.e., plotted as KCOR$(t; t_0)$). This figure is presented as an illustrative application demonstrating estimator behavior on registry data and does not support causal inference. X-axis units are weeks since enrollment.

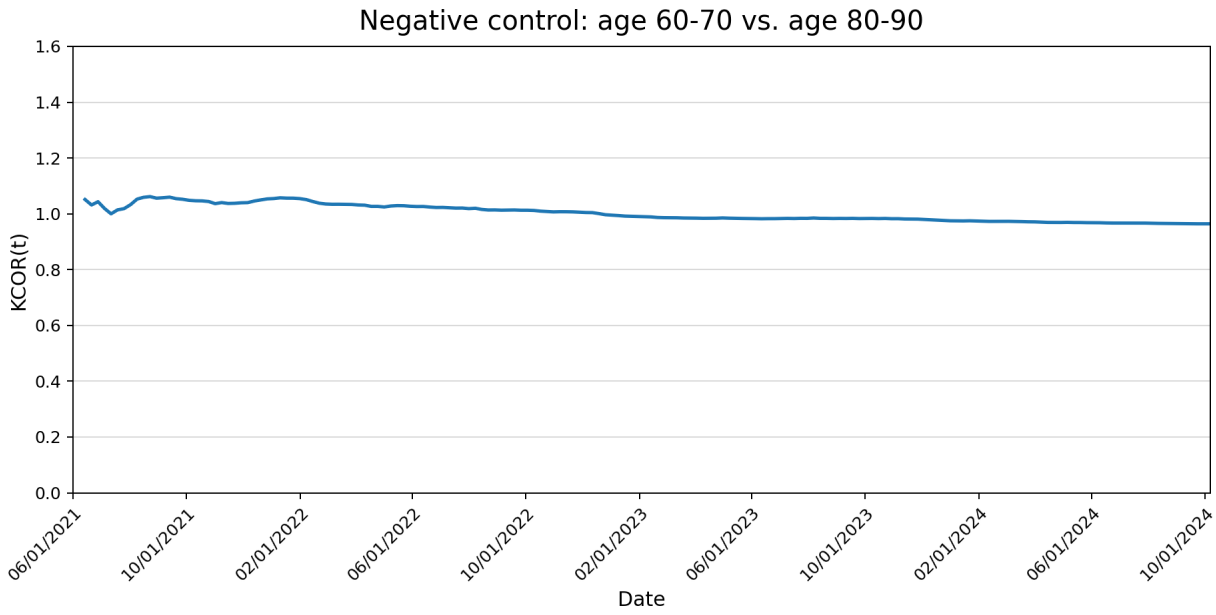


Figure S9: Empirical negative control with approximately 20-year age difference between cohorts. Even under extreme composition differences, KCOR(t) shows little systematic drift, which is consistent with the estimator handling selection-induced curvature in this pseudo-null construction. KCOR curves are anchored at $t_0 = 4$ weeks (i.e., plotted as KCOR$(t; t_0)$). Uncertainty bands (95% bootstrap intervals) are shown. Data source: Czech Republic mortality and vaccination dataset processed into KCOR_CMR aggregated format (negative-control construction; see Supplementary Information, SI). X-axis units are weeks since enrollment.

exemplify this regime: voluntary uptake led to asymmetric selection at enrollment and to visibly different cohort hazard geometry over follow-up. Under the revised estimator, those differences are summarized through fitted $\theta_{0,d}$ values, aligned quiet-window behavior, depletion-normalized cumulative trajectories, and, when invoked, optional NPH diagnostics for the shared amplification exponent α. The Czech application remains illustrative rather than validating; any NPH output should be interpreted as descriptive and model-calibrated rather than as direct biological evidence. While these examples illustrate KCOR's application, the method is general and applies to any retrospective cohort comparison where selection induces differential depletion dynamics.

S6.1.2 Descriptive frailty-normalization behavior in the Czech application

Across examined age strata in the Czech Republic mortality dataset, fitted enrollment-time frailty parameters vary across cohorts, with some estimates close to the weak-identification boundary and others appreciably above zero:

$$\hat{\theta}_{0,d} \geq 0. \tag{3}$$

These values should be interpreted as **descriptive outputs of the revised estimator**, not as validation evidence. They summarize how much depletion curvature the estimator attributes to each cohort at rebased enrollment time under the working model, conditional on the diagnostics for that analysis.

As a consequence, KCOR normalization leaves some cohorts' cumulative hazards nearly unchanged, while substantially increasing the depletion-neutralized baseline cumulative hazard for others. This behavior is consistent with curvature-driven normalization rather than cohort identity. The pattern is visible directly in depletion-neutralized versus observed cumulative hazard plots and is summarized quantitatively in the fitted-parameter logs (see `KCOR_summary.log`).

After frailty normalization, the depletion-neutralized baseline cumulative hazards are approximately linear in event time within diagnostically acceptable quiet windows. Residual deviations from linearity reflect real time-varying

risk—such as seasonality or epidemic waves—rather than selection-induced depletion. This linearization is a diagnostic consistency check under the working model; persistent nonlinearity or parameter instability indicates model stress or quiet-window contamination.

Table S6 summarizes these diagnostic checks across age strata.

Table S6: Diagnostic gate for Czech application: KCOR results reported only where diagnostics pass.

Age band (years)	Quiet window valid	Post-normalization linearity	Parameter stability	KCOR reported
40–49	Yes	Yes	Yes	Yes
50–59	Yes	Yes	Yes	Yes
60–69	Yes	Yes	Yes	Yes
70–79	Yes	Yes	Yes	Yes
80–89	Yes	Yes	Yes	Yes
90–99	Yes	Yes	Yes	Yes
All ages	Yes	Yes	Yes	Yes

In this application, all examined age strata satisfied the prespecified diagnostic criteria; this indicates internal consistency of the working model in this dataset but does not constitute validation of the estimator. KCOR results are not reported for any age stratum where diagnostics indicate non-identifiability.

Interpretation: In this application, some cohorts exhibit larger fitted $\hat{\theta}_{0,d}$ than others, indicating stronger estimated depletion curvature at rebased enrollment time under the working model:

$$\hat{\theta}_{0,d} > 0. \tag{4}$$

for some cohorts, while others remain close to the weak-identification boundary:

$$\hat{\theta}_{0,d} \approx 0. \tag{5}$$

Estimated heterogeneity can appear larger at younger ages because baseline hazards are low, so proportional differences across latent risk strata translate into visibly different short-term hazards before depletion compresses the risk distribution. At older ages, higher baseline hazard and stronger ongoing depletion can reduce the apparent dispersion of remaining risk, yielding smaller fitted $\theta_{0,d}$ even if latent heterogeneity is not literally smaller. These descriptive patterns should not be elevated into a validation criterion, and no single dose-ordering pattern is required by the revised estimator. Table S7 is therefore presented as an empirical summary of fitted parameters rather than as proof that the estimator is correct.

These raw contrasts reflect both selection and depletion effects and are not interpreted causally.

Table S7: Estimated enrollment-time gamma-frailty variance by age band and vaccination status for Czech cohorts enrolled in 2021_24.

Age band (years)	Fitted frailty variance (Dose 0)	Fitted frailty variance (Dose 2)
40–49	16.79	2.66×10^{-6}
50–59	23.02	1.87×10^{-4}
60–69	13.13	7.01×10^{-18}
70–79	6.98	3.46×10^{-17}
80–89	2.97	2.03×10^{-11}
90–99	0.80	8.66×10^{-16}
All ages (full population)	4.98	1.02×10^{-11}

Notes: - The fitted frailty variance quantifies unobserved frailty heterogeneity and selection-induced depletion within cohorts at rebased enrollment time. Near-zero values indicate weak identifiability or effectively linear cumulative hazards under the working model. - Each entry reports a single fitted enrollment-time gamma-frailty variance for the specified age band and vaccination status within the 2021_24 enrollment cohort. - The "All ages (full population)" row corresponds to an independent fit over the full pooled age range, included as a global diagnostic. - Table S8 reports raw outcome contrasts for ages 40+ (YOB $\leq$ 1980) where event counts are stable.

Diagnostic checks: - **Interpretation is conditional on diagnostics:** these values are shown only for strata passing the prespecified diagnostic gate above. - **Weak-identification boundary:** estimates near zero are treated as low-curvature or weak-identification outputs, not as substantive evidence for or against any cohort story. - **Age coherence remains descriptive:** variation across age bands is reported descriptively and may reflect the interaction of baseline hazard, follow-up length, and depletion geometry. - **Stability:** sign reversals, boundary pathologies, or major sensitivity to window perturbation would count against interpretability. - **Falsifiability:** failure of any one of these checks would constitute evidence against model adequacy for that application.

Table S8: Ratio of observed cumulative mortality hazards for unvaccinated (Dose 0) versus fully vaccinated (Dose 2) Czech cohorts enrolled in 2021_24. (Note: the all-ages row reflects aggregation effects and is not directly comparable to age-stratified rows.)

Age band (years)	Dose 0 cumulative hazard	Dose 2 cumulative hazard	Ratio
40–49	0.005260	0.004117	1.2776
50–59	0.014969	0.009582	1.5622
60–69	0.045475	0.023136	1.9655
70–79	0.123097	0.057675	2.1343
80–89	0.307169	0.167345	1.8355
90–99	0.776341	0.517284	1.5008
All ages (full population)	0.023160	0.073323	0.3159

This table reports unadjusted cumulative hazards derived directly from the raw data, prior to any frailty normalization or depletion correction, and is shown to illustrate the magnitude and direction of selection-induced curvature addressed by KCOR.

Values reflect raw cumulative outcome differences prior to KCOR normalization and are not interpreted causally due to cohort non-exchangeability. Cumulative hazards were integrated from cohort enrollment through the end of available follow-up for the 2021_24 enrollment window (through week 2024-16), identically for Dose 0 and Dose 2 cohorts.

S6.1.3 Illustrative application to national registry mortality data

A brief illustrative application is included to demonstrate end-to-end KCOR behavior on real registry mortality data in a setting that minimizes timing-driven shocks and window-tuning sensitivity. Cohorts were enrolled in ISO week 2022-26, and evaluation was restricted to calendar year 2023, yielding a 26-week post-enrollment buffer before slope estimation and a prespecified assessment period. Frailty parameters were estimated using the revised quiet-window settings and diagnostics described above to minimize wave-related hazard variation. This example is intended to illustrate estimator behavior under real-world selection and heterogeneity and does not support causal inference.

Figure S10 shows KCOR(t) trajectories for dose 2 and dose 3 relative to dose 0 for an all-ages analysis. An all-ages analysis is presented as a high-heterogeneity stress test, since aggregation across age induces substantial baseline hazard and frailty variation.

S7. Computational environment and runtime notes

Environment. Python 3.11; key dependencies include numpy, scipy, pandas, and lifelines (for Cox-model comparisons), with plotting via matplotlib.

Compute requirements. The full simulation grid reproduces in approximately 1 hour 26 minutes on a 20-core CPU with 128 GB RAM; smaller subsets reproduce in minutes.

Reproduction. Running `make paper` (or the repository's top-level build command) regenerates all artifacts from a clean checkout.



Figure S10: All-ages stress test: KCOR(t) trajectories comparing dose 2 and dose 3 to dose 0 for cohorts enrolled in ISO week 2022-26 and evaluated over calendar year 2023. KCOR curves are anchored at $t_0 = 4$ weeks (i.e., plotted as KCOR$(t; t_0)$). This figure is presented as an illustrative application demonstrating estimator behavior under extreme heterogeneity and does not support causal inference. X-axis units are weeks since enrollment.